DAILY AI BRIEFING · LEARN ONE THING + WHAT MOVED

Keeping up with the AIs

Memory is the speed limit

Issue #37 | Thu, 05 Nov 2026 | ☕ ~9 min read

⚡ If you only have one minute

1. **Kimi K3.1 Pro's open weights are out**: a 2.8-trillion-parameter model you can download, under a custom licence. Tencent opened its 770B Hy4 the same day.
2. **Claude found two critical Xen bugs** that let one virtual machine read and write another's memory. Patch if you run Xen, XCP-ng or Qubes.
3. **Part 1: AI chips.** Why a chatbot's typing speed is set by memory, not maths, and why Cerebras is so fast.

Today in three numbers

104B
parameters **Kimi K3.1 Pro** actually uses per token, out of 2.8 trillion

21 PB/s
on-chip memory bandwidth of a **Cerebras WSE-3**, vs 3.35 TB/s for an H100

6.5–54.6x
cheaper long tasks when agents **compact their history**, per a new paper

PART 1 Learn one thing

🧠 GPUs, TPUs and inference chips: why Cerebras is fast

4 MIN READ | MODULE 5 · UNDER THE HOOD

When a model writes a reply, the chip spends most of its time fetching weights from memory, not doing maths. So the fastest AI chips are the ones that move data fastest.

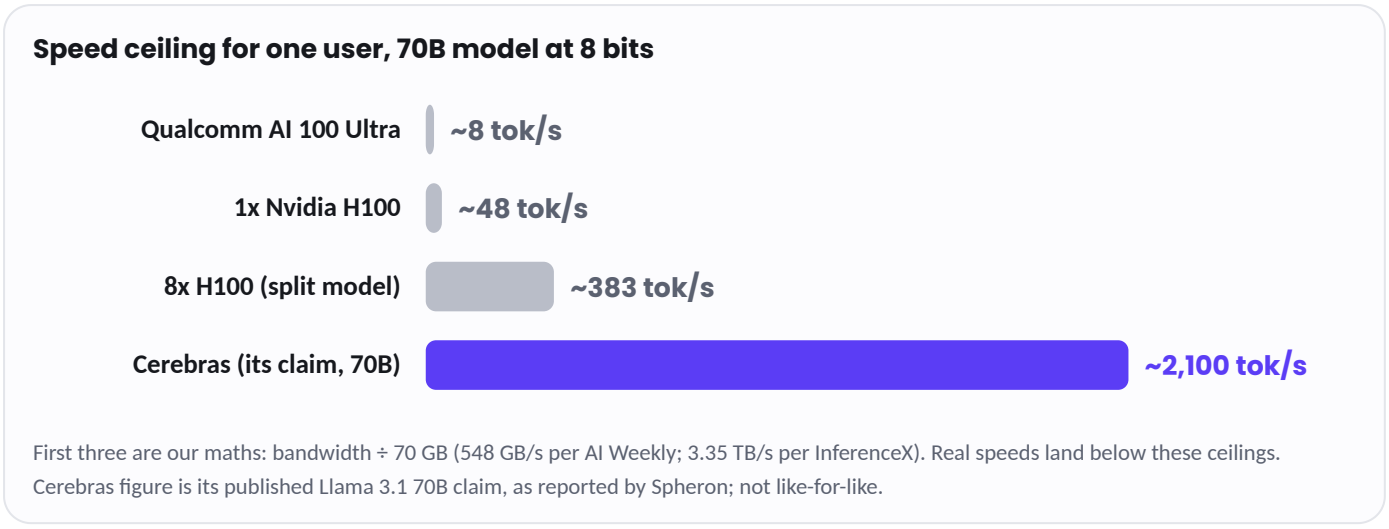
The plain-English version. To produce **each new token**, a model has to touch **every weight it uses**. For a 70B model at 8 bits, that's about **70 GB read from memory for every single word** of the reply. The maths per weight is tiny: one multiply and one add.

A modern GPU can do far more maths than its memory can feed it. An Nvidia H100 does about **989 trillion BF16 operations a second**, but its memory (called **HBM**) delivers "only" **3.35 TB/s**. (InferenceX) So while one person chats, the maths units mostly sit waiting. Engineers call this being **memory-bandwidth-bound**.

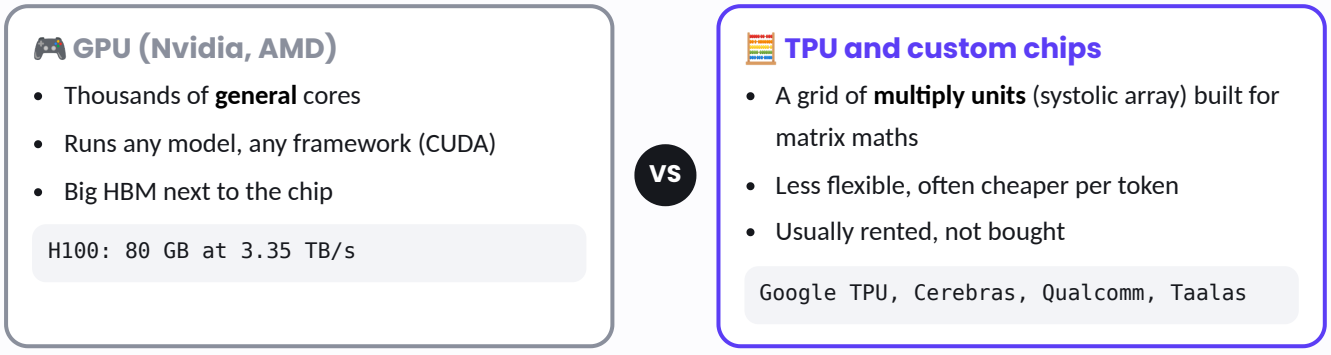
Analogy: a Mumbai dabbawala crew with one handcart. The crew can sort and deliver tiffins at lightning speed, but every tiffin first has to come in on the cart from the kitchen. **Hire faster sorters and nothing changes. Build a wider road to the kitchen and everything does.** Chip designers are mostly building wider roads.



This gives you a handy rule. The most tokens per second one user can get is roughly **memory bandwidth ÷ model size in bytes**. Plug in a few chips and the ranking jumps out.



So why is Cerebras fast? Instead of cutting a silicon wafer into hundreds of chips, it uses **the whole wafer as one chip**: 46,225 mm², about **57 times an H100**. It keeps the model in **44 GB of on-chip SRAM** (the super-fast memory right next to the cores) with **21 PB/s** of bandwidth. (TweakTown) **That's thousands of times the H100's road.** The catch: 44 GB is small, so big models are spread across several wafers.



Why now? Chips are all over this week's news. Qualcomm's **150-watt Cloud AI 100 Ultra** card (128 GB of LPDDR4x at 548 GB/s) hit Hacker News. AMD's filing prices its purchase of inference-chip startup **Taalas at about $2.9B**. And Google is backing Anthropic's **$19B Kentucky lease**, which ties Anthropic to Google-approved **TPUs** there. (AI Weekly)

📖 Jargon buster

- **HBM:** high-bandwidth memory, stacked right beside a GPU. **Big and fast**, but still the bottleneck.
- **SRAM:** memory **on the chip itself**. Much faster, much smaller and pricier per GB.
- **Memory-bandwidth-bound:** speed is limited by **how fast weights arrive**, not how fast you multiply.
- **Batching:** serving many users at once, so **one weight fetch serves everyone**. That's how providers make GPUs pay.

🧪 Try it in 10 minutes

- **No code:** ask the same question to one model on two providers listed on **OpenRouter** and compare tokens per second. Then check which chips each provider says it runs on.
- **Some code:** run the snippet below with **your own laptop's memory bandwidth** (search your chip's name plus "memory bandwidth") and your favourite model's size.

🛠️ For devs: estimate speed before you buy

```
def tok_per_sec_ceiling(params_billion, bits, bandwidth_gb_s):
    model_gb = params_billion * bits / 8      # bytes read per token
    return bandwidth_gb_s / model_gb


for chip, bw in {"Qualcomm AI 100 Ultra": 548,
                 "H100": 3350, "8x H100": 8 * 3350}.items():
    print(f"{chip:22s} {tok_per_sec_ceiling(70, 8, bw):5.0f} tok/s")
# Qualcomm AI 100 Ultra      8 tok/s
# H100                      48 tok/s
# 8x H100                  383 tok/s
```

Gotchas: for **mixture-of-experts** models (Issue #35), use the **active** parameters, not the total. Software matters too: per AI Weekly, an **RTX Spark laptop** ran GPT-OSS-120B at **65.4 tok/s on Windows but 13.9 on Linux**, because of locked power settings.

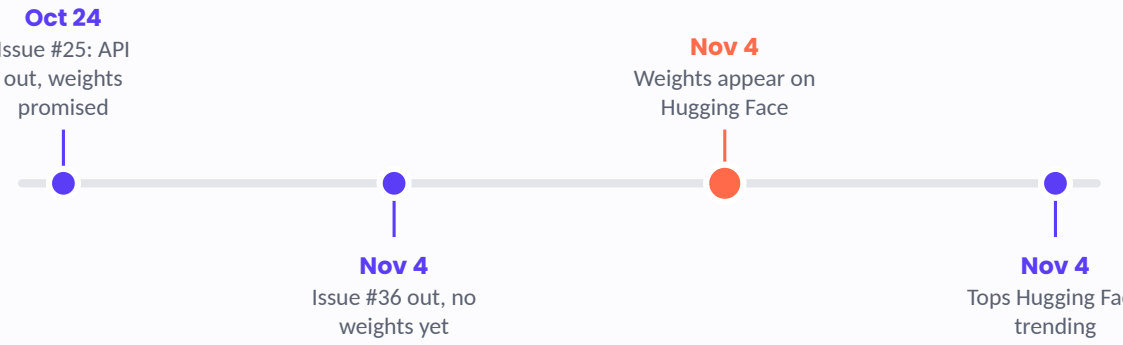
🔗 GO DEEPER

- **Making Deep Learning Go Brrrr From First Principles** (Horace He). Compute vs memory vs overhead, the clearest intro there is.
- **How to Think About TPUs** (Google DeepMind's "How to Scale Your Model"). Systolic arrays and HBM, with diagrams.
- **GPU vs. Wafer-Scale** (MLSys·im). A worked H100 vs Cerebras comparison.
- **TPU architecture** (Google Cloud docs). How TPU chips, cores and pods fit together.

PART 2 What moved in AI

7 min read · Covering the last 24 hours. A giant open model lands on schedule, Claude finds hypervisor bugs, and a paper says your agent doesn't need that million-token window.

Kimi K3.1 Pro: from API to open weights



Top story 1 · Update: Kimi K3.1 Pro's weights land, and Tencent opens Hy4

BIG DEAL OPEN WEIGHTS

What happened. Moonshot kept its promise: **Kimi K3.1 Pro's weights are on Hugging Face**, late on 4 November (we went to press without them yesterday). It's a **2.8-trillion-parameter mixture-of-experts model** that activates **104B per token**, with a **1M-token context** and built-in vision. (AI Weekly) (Hugging Face)

The details that matter:

- **Built for 4-bit:** weights were trained to run in **MXFP4** with 8-bit activations, the "quantization-aware training" from yesterday's Learn.
- **Experts: 896 experts, 16 picked per token** plus 2 shared, across 93 layers.
- **Claimed scores: 65.7 on Terminal-Bench 4.0** and 94.2 on GPQA-Diamond. These are Moonshot's own numbers; **no independent tests yet**.
- **Licence:** a **custom "Kimi K3.1 Pro License"**, not Apache or MIT. Read it before you build a product on it.
- **Same day:** Tencent released **Hy4, a 770B open-weights MoE** with a 1M context and computer-use skills, on Tencent Cloud and OpenRouter, with a **50% launch discount until 31 December**. (AI Weekly)

Why it matters. Today's Learn maths in action: 104B active at about 4 bits is **roughly 52 GB read per token**, and all 2.8T weights (about **1.4 TB at 4 bits**, our maths) must sit in memory. That's a rack of GPUs. **For almost everyone this is an API model**, but the open weights mean many hosts will compete on its price.

WHO SHOULD CARE 🙋 Everyone ●○○○○ 🎨 Vibe coders ●●○○○ 🛠️ Developers ●●●●○

🔥 Hype level ●●●○○

> 👀 **Should you care?** If you use coding agents, yes: watch for **cheaper K3.1 Pro endpoints** on OpenRouter over the next week. Self-hosting is for teams with serious hardware.

Top story 2 · Claude found two critical Xen hypervisor bugs

SECURITY PATCH NOW

What happened. The Xen Project's advisory **XSA-523** credits **Anthropic's Claude** with finding **CVE-2026-102703 and CVE-2026-102704**. Both let a **malicious guest VM read and write other VMs' memory**, including dom0, the control VM that runs the host. (AI Weekly) (The Hacker News)

The details that matter:

- **Where:** Xen's **grant-copy** code, which copies memory between VMs. A shortcut that cached already-checked page numbers **re-checked them wrongly** in some cases.
- **Who's hit: every Xen release from 4.6 on**. There are no patches for 4.6 to 4.16, which are out of support.
- **Fixes:** XCP-ng 8.3 and Qubes OS 4.3 have updates; both need a **host or dom0 restart**.
- **Checked by humans:** Ada Logics validated the findings, and **CodeAnt AI reported the same bugs independently**. The patch carries an "Assisted-by: Claude." line.
- **Unknown:** Anthropic hasn't said **how much a human steered** the search.

Why it matters. A hypervisor is the wall between strangers' VMs on a shared server. **AI models are now finding serious bugs in mature, heavily reviewed software.** That helps defenders, but the same skill helps attackers.

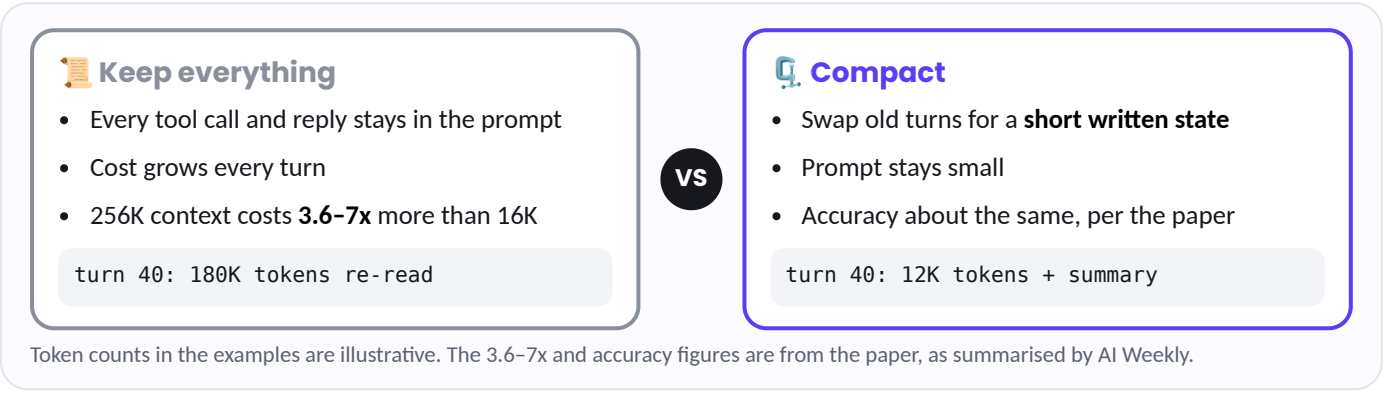
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🔥 Hype level ●●○○○

> 👀 **Should you care?** Only if you run Xen yourself (homelabs on XCP-ng, Qubes laptops, self-hosted servers). Then **patch and reboot today**.

Top story 3 · "You don't need long context," says a new paper

TRY IT FOR DEVS

What happened. A new paper tests whether agents really need giant context windows. It finds that **compacting the history** (replacing old turns with a short summary) keeps accuracy within **−3.7 to +5.6 points** of keeping everything, while cutting costs **6.5 to 54.6 times** on long tasks. (AI Weekly) (Hugging Face Papers)



Token counts in the examples are illustrative. The 3.6–7x and accuracy figures are from the paper, as summarised by AI Weekly.

The details that matter:

- **Two tricks:** compact once a conversation ends (**EoCC**), or after **every single action** (**SAC**).

- **Tested widely:** SWE-bench, Terminal-Bench 2.1 and LongBench v2, on models including Claude Opus 4.5, GLM 5.2, Kimi K2.6 and Gemma 4.
- **A new idea:** "**semantic state space models**", which carry a compact plain-English state forward instead of the whole transcript.

Why it matters. Long context is **expensive and slow** (see Part 1: more tokens, more memory traffic). If a summary does the job, **your agent bill could shrink a lot** without a model change.

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🔥 Hype level ●●○○○

🛠️ For devs: a cheap compaction loop

Every N tool calls, ask a small model to rewrite the history as **goal, decisions made, open TODOs, key file paths**, then replace the old turns with that note. Keep the **last 2–3 turns verbatim**. Run your evals (Issue #26) with and without it before switching.

⚡ Quick hits

🧪 0.1% poisoned data plants an agent backdoor.

Anthropic, UC Berkeley and others show poisoned pretraining text made Qwen3 agents run an attacker's script on **92.7% of triggered tasks**, surviving fine-tuning and RL. (Kill The AI) (arXiv)

🔐 OpenAI hires the firm that got into Anthropic.

Cathedral Systems will test OpenAI's defences; the same day, Reuters reported years of internal security warnings. **OpenAI disputes** neglecting security. (The Neuron) (OpenAI)

🤝 Meta buys Conductor.

The app runs **Claude Code, Codex and Cursor agents in parallel**. Write access ends **13 Nov**; its cloud shuts **20 Nov**. Export your work. (AI Weekly)

💸 Revolut cuts AI coding costs ~90%.

It made **self-hosted GLM models the default** and keeps commercial models for when it needs them. (The Neuron, via The Register)

👁️ Gemini trains on paid users' chats.

With **Keep Activity** on (the default), saved chats from personal accounts, **paid tiers included**, train Google's models. API and Workspace are exempt. (AI Weekly)

🍎 Apple trims its AI allowance.

Some engineers' daily AI-tool cap fell from **$300 to $200**, and some teams may drop to $100, per Bloomberg. (AI Weekly)

🗳️ Voters push back on data centres.

Ohio towns voted on **21 data-centre measures**, and many passed bans or limits; a statewide poll found **55% opposed** to new construction. (AI Weekly)

🗓️ Coming up.

Ollie shuts **6 Nov**. Sonos 27voice waitlist opens **6 Nov**. US lame-duck session **9 Nov**. Anthropic usage policy **12 Nov**. Conductor read-only **13 Nov**.

🎯 One thing to try today: find your own speed limit

Look up your laptop's **memory bandwidth** (an M-series Mac or recent PC is listed on the spec sheet). Divide by the size of a model file you run in Ollama or LM Studio. Then time a real reply. **If you're close to the ceiling, only a smaller model, or more bandwidth, will make it faster.**

🧠 Pop quiz: 30 seconds, no Googling

1. What usually limits how fast one user's reply is generated?

- A Maths speed
- B Memory bandwidth
- C Internet speed

2. Where does Cerebras keep the model while it runs?

- A On-chip SRAM
- B Hard disk
- C Cloud storage

3. What did the long-context paper replace old turns with?

- A Nothing
- B A short summary
- C Images

Answers (no peeking): 1. B: every weight is read once per token · 2. A: 44 GB of SRAM on one wafer · 3. B: compaction keeps a compact state

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